TANTA UNIVERSITY
Faculty of Computers and Information

CS433
Selected Topics in Artificial Intelligence

Question Bank 3 – Solved
Week 1: Introduction – What Is Artificial Intelligence?

Contents
Part A: 30 Multiple-Choice Questions
Part B: 10 True/False Questions
Every question includes the original choices, the correct answer, an English explanation, an Arabic explanation, and a lecture reference.

Prepared for study and revision

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1 Part A – Multiple Choice

A1

Question: Which of the four approaches to AI does the textbook adopt?

(a) Thinking humanly

(b) Acting humanly

(c) Acting rationally

(d) Thinking rationally

Correct answer

(c) Acting rationally.

Explanation (English): The textbook adopts the rational-agent approach. It focuses on selecting actions that achieve the best outcome, or the best expected outcome under uncertainty, rather than imitating human thought or behaviour.

الشرح بالعربي: الكتاب بيتبنى منهج التصرف العقلاني؛ يعني بناء وكيل ذكي يختار الفعل اللي يحقق أفضل نتيجة ممكنة، أو أفضل نتيجة متوقعة لو فيه عدم يقين، مش شرط يقلد طريقة تفكير الإنسان.

Reference: CS433-W01-Lecture-Notes, §1.1.4, pp. 3–4.

A2

Question: Which statement best describes the relationship between AI and machine learning?

(a) Machine learning is a subfield of AI that studies how to improve performance from experience.

(b) AI is a subfield of machine learning.

(c) AI and machine learning are two names for the same field.

(d) Every AI system, including route planners and theorem provers, uses machine learning.

Correct answer

(a) Machine learning is a subfield of AI that studies how to improve performance from experience.

Explanation (English): Machine learning is part of AI, not a synonym for it. Some AI systems learn from data, while others rely on search, planning, or logical reasoning without machine learning.

الشرح بالعربي: التعلم الآلي جزء من الذكاء الاصطناعي، ويختص بتحسين أداء النظام من خلال الخبرة

والبيانات. لكن مش كل نظام ذكاء اصطناعي لازم يستخدم Machine Learning، زي مخطط الطرق أو مبرهن النظريات.

Reference: CS433-W01-Lecture-Notes, introductory note before §1.1, p. 1.

A3

Question: Which two questions separate the four historical definitions of AI?

(a) Is AI built in hardware or software, and is it fast or accurate?

(b) Is AI symbolic or neural, and is it supervised or unsupervised?

(c) Is AI used in industry or research, and is it cheap or expensive?

(d) Is intelligence about internal thought or external behaviour, and is it measured against human performance or rationality?

Correct answer

(d) Internal thought or external behaviour; human performance or rationality.

Explanation (English): The four approaches are formed by combining two distinctions: thinking versus acting, and human-like performance versus ideal rationality.

الشرح بالعربي: التقسيم مبني على سؤالين: هل بنركز على التفكير الداخلي ولا السلوك الخارجي؟ وهل بنقارن أداء الآلة بالإنسان ولا بالمثالية والعقلانية؟ ومن دمج السؤالين بيطلعوا الأربع اتجاهات.

Reference: CS433-W01-Lecture-Notes, §1.1, pp. 1–2.

A4

Question: What does the total Turing test require in addition to the standard Turing test?

(a) Solving problems faster than a human

(b) Interaction with objects and people in the physical world

(c) Proving theorems from Principia Mathematica

(d) Passing the test without any written questions

Correct answer

(b) Interaction with objects and people in the physical world.

Explanation (English): The standard test is based on written interaction with a human interrogator. The total Turing test adds perception and physical interaction, including vision and robotics.

الشرح بالعربي: اختبار تورينج العادي بيعتمد على محادثة مكتوبة، بحيث المحكّم مايعرفش يميز هل بيرد عليه إنسان ولا آلة. الاختبار الكلي بيضيف التعامل مع العالم الحقيقي، زي الرؤية والتفاعل مع الأشياء والأشخاص.

Reference: CS433-W01-Lecture-Notes, §1.1.1, pp. 2–3.

A5

Question: What point does the book make with the example of artificial flight?

(a) AI should study the underlying principles of intelligence rather than copy humans.

(b) AI should copy the human brain as closely as possible.

(c) Machines will never be able to think, just as they cannot fly like birds.

(d) The Turing test should be replaced by a test that fools pigeons.

Correct answer

(a) AI should study the underlying principles of intelligence rather than copy humans.

Explanation (English): Artificial flight succeeded by applying aerodynamics, not by building machines that imitate birds exactly. Similarly, AI should understand and apply principles of intelligence instead of merely copying people.

الشرح بالعربي: الطيران الصناعي نجح لما المهندسين درسوا قوانين الديناميكا الهوائية، مش لما حاولوا يعملوا آلة بتقلد الطيور في كل حركة. ونفس الفكرة في الذكاء الاصطناعي: ندرس مبادئ الذكاء بدل تقليد الإنسان حرفيًا.

Reference: CS433-W01-Lecture-Notes, §1.1.1, p. 3.

A6

Question: Newell and Simon's General Problem Solver (GPS) was judged not only on solving problems but also on:

(a) how few lines of code it needed

(b) whether it could fool a human interrogator

(c) whether the sequence and timing of its reasoning steps matched human subjects

(d) whether it could learn from experience

Correct answer

(c) Whether the sequence and timing of its reasoning steps matched human subjects.

Explanation (English): GPS was studied as a cognitive model. Researchers compared its reasoning steps and their timing with those observed in human problem-solving

experiments.

الشرح بالعربي: برنامج GPS ماكانش التقييم بتاعه بس هل يحل المسألة ولا لأ، كمان كانوا بيقارنوا خطوات التفكير وترتيبها والوقت اللي بتاخده بخطوات وتوقيت تفكير البشر. وده مثال على Cognitive Modeling.

Reference: CS433-W01-Lecture-Notes, §1.1.2, p. 3.

A7

Question: Logic needs certain knowledge, which is rarely available. According to the notes, what fills this gap?

(a) Syllogisms

(b) Introspection

(c) Brain imaging

(d) Probability

Correct answer

(d) Probability.

Explanation (English): Logical reasoning often requires certain premises, but real-world knowledge is frequently uncertain. Probability provides a way to reason when certainty is unavailable.

الشرح بالعربي: المنطق التقليدي محتاج معلومات مؤكدة، لكن في الواقع كتير من المعلومات بتكون غير مؤكدة. الاحتمالات بتساعدنا نتعامل مع عدم اليقين ونستنتج بشكل مناسب.

Reference: CS433-W01-Lecture-Notes, §1.1.3, p. 3.

A8

Question: A rational agent is one that acts so as to achieve:

(a) the outcome a human would choose

(b) the best outcome or, when there is uncertainty, the best expected outcome

(c) a correct logical inference in every situation

(d) the fastest possible response

Correct answer

Second listed choice: the best outcome or, under uncertainty, the best expected outcome.

Explanation (English): This is the definition of a rational agent in the lecture. Note: the supplied question bank prints the label “(a)” twice for its first two choices; the correct answer is the second listed choice, whose label appears to be a typographical error.

الشرح بالعربي: الوكيل العقلاني مش لازم يختار نفس الاختيار اللي الإنسان هيختاره، ومش شرط يعمل استنتاج منطقي في كل موقف. هو يختار الفعل اللي يحقق أفضل نتيجة، ولو فيه احتمالات وعدم يقين يختار أفضل نتيجة متوقعة. ملحوظة: البنك الأصلي مكرر فيه الرمز (a) لأول اختيارين، فالإجابة هي الاختيار الثاني في الترتيب.

Reference: CS433-W01-Lecture-Notes, §1.1.4, p. 4.

A9

Question: The standard model appears in many fields under different names. In statistics, the agent:

(a) minimises a cost function

(b) maximises a sum of rewards

(c) minimises a loss function

(d) maximises utility

Correct answer

(c) Minimises a loss function.

Explanation (English): The lecture maps the standard model to different fields: control theory minimises cost, operations research maximises rewards, statistics minimises loss, and economics maximises utility or social welfare.

الشرح بالعربي: في الإحصاء، الهدف بيكون تقليل دالة الخسارة (Loss Function). وافتكر الفرق: Control Theory بتقلل Cost، وResearch Operations بتزود Rewards، وEconomics بتعظم Utility.

Reference: CS433-W01-Lecture-Notes, §1.1.4, p. 4.

A10

Question: What is the first refinement of the standard model, called limited rationality?

(a) Agents must act appropriately when there is not enough time to do every computation.

(b) Agents must be uncertain about the human objective.

(c) Agents must imitate human reasoning steps.

(d) Agents must use only logic, never probability.

Correct answer

(a) Agents must act appropriately when there is not enough time to do every computation.

Explanation (English): Perfect rationality may require more computation than is practical. Limited rationality recognises computational limits and requires an agent to make appropriate decisions with the time and resources available.

الشرح بالعربي: العقلانية الكاملة معناها إن النظام يحسب كل حاجة لحد ما يوصل لأفضل قرار مثالي، وده ممكن يحتاج وقت وحسابات ضخمة. Rationality Limited معناها إنه ياخد قرار مناسب بالإمكانيات والوقت المتاحين.

Reference: CS433-W01-Lecture-Notes, §1.1.4, p. 4.

A11

Question: What is the value alignment problem?

(a) Making sure an agent's computations finish in time

(b) Choosing between expected money and expected utility

(c) Matching a program's reasoning steps with those of humans

(d) Making sure the values or objectives put into the machine are aligned with those of the human

Correct answer

(d) Aligning the machine's values or objectives with human values.

Explanation (English): The value alignment problem arises because a machine may optimise an objective that does not accurately represent what humans really want. The objective must be aligned with human preferences.

الشرح بالعربي: مشكلة توافق القيم معناها إن الهدف اللي بندخله للآلة لازم يكون متوافق مع اللي الإنسان عايزه فعلًا. لأن الآلة ممكن تنفذ الهدف حرفيًا وبشكل ممتاز، لكن النتيجة تكون عكس رغبتنا الحقيقية.

Reference: CS433-W01-Lecture-Notes, §1.1.5, pp. 4–5.

A12

Question: According to the notes, a machine that knows it does not know the full human objective has an incentive to:

(a) pursue its own objective as fast as possible

(b) act cautiously, ask permission, learn our preferences and defer to human control

(c) hijack extra computing power to find the objective

(d) switch itself off permanently

Correct answer

(b) Act cautiously, ask permission, learn our preferences and defer to human control.

Explanation (English): If the machine is uncertain about the human objective, it has reason to seek clarification and allow humans to guide or correct it instead of blindly pursuing a possibly incorrect fixed objective.

الشرح بالعربي: لما الآلة تبقى عارفة إنها مش فاهمة هدف الإنسان بالكامل، المفروض تتصرف بحذر، وتسأل قبل القرارات المهمة، وتتعلم تفضيلاتنا، وتسيب للإنسان سلطة التحكم والتوجيه.

Reference: CS433-W01-Lecture-Notes, §1.1.5, p. 5.

A13

Question: The word “algorithm” comes from the name of:

(a) Aristotle

(b) al-Khwarizmi

(c) Boole

(d) Gödel

Correct answer

(b) al-Khwarizmi.

Explanation (English): The word algorithm is derived from the Latinized form of the name of the mathematician al-Khwarizmi.

الشرح بالعربي: كلمة Algorithm أصلها مرتبط باسم العالم المسلم الخوارزمي، والاسم اتنقل للاتينية بصيغة قريبة من كلمة Algorithm.

Reference: CS433-W01-Lecture-Notes, §1.2 (Mathematics), p. 6.

A14

Question: Which is the correct order of the three steps of Craik’s knowledge-based agent (1943)?

(a) Act, then observe, then remember

(b) Manipulate the representation, then sense the stimulus, then act

(c) Translate the stimulus into an internal representation, manipulate it to derive new representations, then translate them back into action

(d) Translate action into a stimulus, then build a representation, then stop

Correct answer

(c) Translate the stimulus into an internal representation, manipulate it, then translate it into action.

Explanation (English): Craik’s three stages are: represent the stimulus internally, manipulate the representation to derive new ones, and convert the resulting representation into action.

الشرح بالعربي: كرايك وضح إن الوكيل اللي بيفكر بيمر بثلاث خطوات: أولًا يحول المؤثر اللي استقبله لتمثيل داخلي، ثانيًا يعالج التمثيل ده ويستنتج معلومات جديدة، وثالثًا يحول النتيجة لفعل أو تصرف. يعني: Act. ← Reason ← Represent

Reference: CS433-W01-Lecture-Notes, §1.2, pp. 7–8.

A15

Question: You can choose (A) 1,000,000 EGP for certain or (B) a coin flip giving 3,000,000 EGP or nothing. What is the expected money of option B?

(a) 1,500,000 EGP

(b) 1,000,000 EGP

(c) 3,000,000 EGP

(d) 750,000 EGP

Correct answer

(a) 1,500,000 EGP.

Explanation (English): A fair coin gives each outcome probability 0.5. Therefore,

$$E(B) = 0.5(3{,}000{,}000) + 0.5(0) = 1{,}500{,}000 \text{ EGP}.$$

الشرح بالعربي: بما إن العملة عادلة، احتمال تكسب 3 مليون هو 50% واحتمال تكسب صفر هو 50%.
القيمة المتوقعة = (0.5 × 3 مليون) + (0.5 × صفر) = 1.5 مليون جنيه.

Reference: CS433-W01-Lecture-Notes, §1.2, discussion of utility and expected value, pp. 8–9.

A16

Question: In the same choice, let $U(0) = 0$, $U(1M) = 0.8$ and $U(3M) = 1$. Which option does a rational agent choose?

(a) B, because its expected money is higher

(b) B, because $EU(B) = 1$

(c) Either, because the expected utilities are equal

(d) A, because $EU(A) = 0.8$ is greater than $EU(B) = 0.5$

Correct answer

(d) A, because $EU(A) = 0.8 > EU(B) = 0.5$.

Explanation (English): For A, the utility is certain: $EU(A) = 0.8$. For B,

$$EU(B) = 0.5(1) + 0.5(0) = 0.5.$$

Since $0.8 > 0.5$, the rational choice under the given utility function is A.

الشرح بالعربي: هنا مش بنقارن الفلوس المتوقعة، لكن بنقارن المنفعة المتوقعة. الاختيار A مضمونة منفعته 8.0. أما B فمنفعته المتوقعة = 50% × 1 + 50% × 0 = 5.0. إذن A أفضل لأن 8.0 أكبر من 5.0.

Reference: CS433-W01-Lecture-Notes, §1.2, utility and decision theory, pp. 8–9.

A17

Question: A problem is called intractable if:

(a) no program can solve it at all

(b) it cannot be written in first-order logic

(c) its solution time grows exponentially with the instance size

(d) it needs a parallel computer

Correct answer

(c) Its solution time grows exponentially with the instance size.

Explanation (English): Intractability concerns computational cost, not whether a solution exists. Exponential growth makes solving large real-world instances impractical.

الشرح بالعربي: المشكلة Intractable مش معناها إن مفيش برنامج يقدر يحلها، لكن معناها إن وقت الحل بيزيد بشكل أُسّي مع زيادة حجم المشكلة، وده بيخلي حل الحالات الكبيرة غير عملي.

Reference: CS433-W01-Lecture-Notes, §1.2, mathematics and tractability, pp. 6 and 9.

A18

Question: The brain's cycle time is 10^{-3} s and a computer's is 10^{-9} s. How many times faster per step is the computer?

(a) 10^{6}

(b) 10^{3}

(c) 10^{7}

(d) 10^{12}

Correct answer

(a) 10^{6} **times.**

Explanation (English): The speed ratio is the brain cycle time divided by the computer cycle time:

$$\frac{10^{-3}}{10^{-9}} = 10^{-3-(-9)} = 10^{6}.$$

الشرح بالعربي: بنقسم زمن خطوة المخ على زمن خطوة الكمبيوتر: 10 أس سالب 3 على 10 أس سالب 9.

عند القسمة بنطرح الأسس، فيكون الناتج 10 أس 6؛ يعني الكمبيوتر أسرع بمليون مرة في زمن الخطوة الواحدة.

Reference: CS433-W01-Lecture-Notes, §1.2, neuroscience and computer engineering discussion, pp. 7–10.

A19

Question: From 2012 to 2018 the compute used to train top ML systems grew 300,000-fold. About how many doublings is that?

(a) About 5

(b) About 18

(c) About 100

(d) About 3.4

Correct answer

(b) About 18.

Explanation (English): The number of doublings is $\log_2(300{,}000)$:

$$\log_2(300{,}000) \approx 18.2.$$

So the increase is about 18 doublings.

الشرح بالعربي: عدد مرات التضاعف بنحسبه باستخدام لوغاريتم 300 ألف للأساس 2. الناتج تقريبًا 18.2، يعني حوالي 18 مرة تضاعف.

Reference: CS433-W01-Lecture-Notes, §1.2, hardware for AI, p. 10.

A20

Question: What did McCulloch and Pitts (1943) do?

(a) They built SNARC, the first neural-network computer.

(b) They proposed a rule for strengthening connections between neurons.

(c) They proved that a perceptron cannot learn XOR.

(d) They modelled neurons as “on” or “off” and showed networks of them can compute any computable function.

Correct answer

(d) They modelled neurons as “on” or “off” and showed networks of them can compute any computable function.

Explanation (English): McCulloch and Pitts introduced a simplified binary neuron model and demonstrated the computational power of networks built from such units.

الشرح بالعربي: ماكولوتش وبيتس قدموا نموذج مبسط للعصبون بيكون يا شغال (1) يا مطفي (0)، ووضحوا إن شبكات من الوحدات دي تقدر تنفذ أي دالة قابلة للحساب.

Reference: CS433-W01-Lecture-Notes, §1.3.1 and lab activity, pp. 11 and 17.

A21

Question: What was the main outcome of the Dartmouth workshop of summer 1956, organised by McCarthy?

(a) It gave the field of AI its name.

(b) It produced several major breakthroughs.

(c) It introduced the Turing test.

(d) It ended AI funding in Britain.

Correct answer

(a) It gave the field of AI its name.

Explanation (English): The Dartmouth workshop is historically associated with the naming of the field of artificial intelligence. It did not itself produce the major breakthroughs sometimes associated with later AI development.

الشرح بالعربي: ورشة دارتموث سنة 1956 مهمة لأنها ارتبطت بإطلاق اسم Intelligence Artfiicial على المجال، وكانت نقطة تأسيسية في تاريخ الذكاء الاصطناعي.

Reference: CS433-W01-Lecture-Notes, §1.3.1, history of AI, pp. 11–12.

A22

Question: Samuel's checkers program learned to play better than its creator using what we now call:

(a) expert systems

(b) Bayesian networks

(c) reinforcement learning

(d) resolution theorem proving

Correct answer

(c) Reinforcement learning.

Explanation (English): Samuel's checkers program improved through experience and feedback from playing games. This is an early example of what is now called reinforcement learning.

الشرح بالعربي: برنامج صموئيل للعبة الداما كان بيتحسن من خلال اللعب والخبرة والتغذية الراجعة. النوع ده من التعلم بنسميه حاليًا Reinforcement Learning.

Reference: CS433-W01-Lecture-Notes, §1.3, history of AI, pp. 12–13.

A23

Question: What did Minsky and Papert's book *Perceptrons* (1969) show?

(a) A perceptron can represent any function.

(b) A two-input perceptron cannot learn to tell whether its inputs differ (XOR).

(c) Back-propagation cannot train multilayer networks.

(d) Perceptrons learn faster than expert systems.

Correct answer

(b) A two-input perceptron cannot learn XOR.

Explanation (English): A single perceptron cannot represent XOR because XOR is not linearly separable. The limitation concerns a single unit, not the impossibility of using multilayer networks.

الشرح بالعربي: كتاب Perceptrons وضّح إن وحدة Perceptron واحدة مش قادرة تنفذ XOR، لأن الحالات اللي ناتجها 1 مش ممكن نفصلها عن الحالات اللي ناتجها 0 بخط مستقيم واحد. ده لا يعني إن الشبكات متعددة الطبقات مستحيلة.

Reference: CS433-W01-Lecture-Notes, §1.3 and McCulloch–Pitts lab, pp. 13 and 17–18.

A24

Question: DENDRAL's ketone rule requires $x_1 + x_2 = M + 28$. For a molecule with $M = 120$ and a peak at $x_1 = 80$, which value of x_2 satisfies this condition?

(a) 40

(b) 52

(c) 92

(d) 68

Correct answer

(d) 68.

Explanation (English): Substitute the given values:

$$x_1 + x_2 = M + 28 \Rightarrow 80 + x_2 = 120 + 28 = 148$$

$$x_2 = 148 - 80 = 68.$$

الشرح بالعربي: نعوض في القانون: 80 + قيمة x2 = 120 + 28. الطرف اليمين يساوي 148، إذن x2 = 148 - 80 = 68.

Reference: CS433-W01-Lecture-Notes, §1.3, knowledge-based and expert systems history, pp. 13–14.

A25

Question: What did Banko and Brill (2001) find?

(a) Unlabelled data cannot improve accuracy.

(b) Better algorithms always beat more data.

(c) Increasing the data by two or three orders of magnitude outweighs any improvement from tweaking the algorithm.

(d) Filling holes in photos needs only thousands of images.

Correct answer

(c) Increasing the data by two or three orders of magnitude can outweigh algorithm tweaking.

Explanation (English): Their finding highlighted the importance of data scale: substantially increasing the amount of data could improve performance more than making smaller changes to the learning algorithm.

الشرح بالعربي: Banko وBrill لقوا إن زيادة حجم البيانات بشكل ضخم، بمقدار 100 أو 1000 مرة تقريبًا، ممكن تحسن الأداء أكتر من مجرد تعديلات بسيطة في الخوارزمية نفسها.

Reference: CS433-W01-Lecture-Notes, §1.3, data and machine learning, pp. 13–14.

A26

Question: According to the notes, deep learning is machine learning with multiple layers of simple, adjustable computing elements. When did its take-off come?

(a) 2011, first in speech and then in vision

(b) 1986, with Parallel Distributed Processing

(c) 1997, with Deep Blue

(d) 2022, with ChatGPT

Correct answer

(a) 2011, first in speech and then in vision.

Explanation (English): The lecture describes the modern take-off of deep learning around 2011, with early major progress in speech recognition followed by computer vision.

الشرح بالعربي: الانطلاقة الحديثة للتعلم العميق كانت حوالي 2011، ظهر تأثيره الأول بشكل واضح في التعرف على الكلام، وبعد كده في مجال الرؤية الحاسوبية.

Reference: CS433-W01-Lecture-Notes, §1.3, deep learning era, pp. 14–15.

A27

Question: According to the book, why are parole or loan decisions made by AI sometimes biased by race or gender?

(a) Because the hardware is too slow

(b) Because the systems use logic instead of probability

(c) Because the decisions are always made by humans

(d) Because the data reflect bias in society

Correct answer

(d) Because the data reflect bias in society.

Explanation (English): AI systems trained on historical or social data can reproduce existing patterns of discrimination. The bias may come from the data rather than from the hardware or the use of a particular reasoning method.

الشرح بالعربي: لو البيانات اللي اتدرب عليها النظام فيها تحيز موجود أصلًا في المجتمع، زي تمييز ضد فئة معينة، فالذكاء الاصطناعي ممكن يتعلم التحيز ده ويكرره في قرارات القروض أو الإفراج المشروط.

Reference: CS433-W01-Lecture-Notes, §1.5, near-term risks, p. 16.

A28

Question: What does the King Midas problem describe?

(a) A more intelligent species leaving us without control over our future

(b) A machine giving us exactly what we asked for, not what we wanted

(c) An AI industry that grows from millions to billions of dollars

(d) A machine that cannot be switched off because it is too slow

Correct answer

(b) A machine giving us exactly what we asked for, not what we wanted.

Explanation (English): The King Midas story illustrates the danger of specifying an objective incorrectly: the system fulfils the literal request, even when the result conflicts with the human's real intentions.

الشرح بالعربي: الملك ميداس طلب إن كل حاجة يلمسها تتحول لذهب، وده حصل فعلًا، لكنه اكتشف إن

الأكل والشرب وأهله كمان بيتحولوا لذهب. المشكلة إن التنفيذ حقق الطلب الحرفي، لكنه ماحققش اللي هو كان عايزه فعلًا.

Reference: CS433-W01-Lecture-Notes, §1.5, long-term risks and control, p. 17.

A29

Question: In the lab, a McCulloch–Pitts unit fires when $\sum wx \geq$ threshold. For the unit with weights $(-1, -1)$ and threshold -1, what is the output for inputs $(1, 1)$, and which gate is it?

(a) 1; it is AND

(b) 1; it is OR

(c) 0; it is NAND

(d) 0; it is NOT

Correct answer

(c) 0; it is NAND.

Explanation (English): For input $(1, 1)$, the weighted sum is

$$(-1)(1) + (-1)(1) = -2.$$

Since $-2 \not\geq -1$, the unit outputs 0. For inputs 00, 01, 10, and 11, the outputs are 1, 1, 1, and 0 respectively, which is NAND.

الشرح بالعربي: لما المدخلين = 1 و1، المجموع = -1 -1 = -2. وبما إن -2 أصغر من Threshold اللي هو -1، الوحدة مش هتشتغل والناتج 0. وباختبار باقي الحالات بنلاقي النواتج 1،1،1،0؛ وده جدول بوابة NAND.

Reference: CS433-W01-Lecture-Notes, McCulloch–Pitts lab activity, pp. 17–18.

A30

Question: In the lab, how is XOR built from two layers of units?

(a) $XOR(a, b) = AND(OR(a, b), NAND(a, b))$

(b) $XOR(a, b) = OR(AND(a, b), NAND(a, b))$

(c) $XOR(a, b) = AND(OR(a, b), AND(a, b))$

(d) $XOR(a, b) = NOT(OR(a, b))$

Correct answer

(a) $XOR(a, b) = AND(OR(a, b), NAND(a, b))$**.**

Explanation (English): The OR unit is 1 when at least one input is 1, while NAND is 1 except when both inputs are 1. ANDing these two outputs gives 1 exactly when the inputs differ.

الشرح بالعربي: بنحسب OR وNAND لنفس المدخلين، وبعد كده بندخل الناتجين على AND. لو المدخلين مختلفين، OR = 1 وNAND = 1، فالناتج النهائي 1. ولو المدخلين متساويين، الناتج النهائي 0. وده بالضبط XOR.

Reference: CS433-W01-Lecture-Notes, McCulloch–Pitts lab activity, pp. 17–18.

2 Part B – True or False

B1

Statement: AI researchers have spent a great deal of effort trying to pass the Turing test.

Answer: False

The lecture explicitly says that AI researchers have spent little effort trying to pass the Turing test because they consider studying the underlying principles of intelligence more important than imitation.
Correction: AI researchers have spent little effort trying to pass the Turing test.

الشرح بالعربي: العبارة غلط، المحاضرة بتقول إن الباحثين بذلوا مجهود قليل في محاولة اجتياز اختبار تورينج، لأن الأهم عندهم فهم مبادئ الذكاء نفسها، مش مجرد تقليد الإنسان.

Reference: CS433-W01-Lecture-Notes, §1.1.1, p. 3.

B2

Statement: For a self-driving car, "reach the destination safely" taken strictly would mean staying in the garage, because every road carries some risk.

Answer: True

The lecture uses this example to show that a literal safety-only objective can prevent the car from moving at all. The objective must balance safety with progress and other human preferences.

الشرح بالعربي: العبارة صح. لو العربية هتفسر الأمان بشكل حرفي جدًا، فبما إن كل طريق فيه نسبة خطر، يبقى الحل الأكثر أمانًا إنها تفضل في الجراج وما تتحركش. وده يوضح أهمية تحديد الهدف بشكل يعبر عن رغبتنا الحقيقية.

Reference: CS433-W01-Lecture-Notes, §1.1.5, pp. 4–5.

B3

Statement: Daniel Bernoulli (1738) introduced utility, the subjective value of an outcome, with diminishing marginal utility of money.

Answer: True

Bernoulli's contribution is associated with utility as a subjective measure of value. Diminishing marginal utility means that an additional unit of money generally has less impact as wealth increases.

الشرح بالعربي: العبارة صح. برنولي قدّم فكرة المنفعة (Utility)، وهي القيمة أو الرضا اللي الشخص يحصل عليه من نتيجة معينة، مش مجرد قيمتها المالية. والمنفعة الحدية للنقود بتقل غالبًا كل ما الثروة تزيد.

Reference: CS433-W01-Lecture-Notes, §1.2, economics and utility, pp. 8–9.

B4

Statement: Back-propagation for multilayer networks was first invented in the mid-1980s.

Answer: False

The mid-1980s are associated with the popularisation and influential use of back-propagation for training multilayer networks, not its first invention.
Correction: Back-propagation became widely influential in the mid-1980s; it was not first invented then.

الشرح بالعربي: العبارة غلط بسبب كلمة first. invented منتصف الثمانينيات ارتبط بانتشار واستخدام Back-propagation بشكل مؤثر في تدريب الشبكات متعددة الطبقات، لكن مش معناه إنها اتاخترعت لأول مرة في الفترة دي.

Reference: CS433-W01-Lecture-Notes, §1.3, history of neural networks and AI winter, pp. 13–14.

B5

Statement: R1/XCON was the first successful knowledge-intensive system.

Answer: False

DENDRAL is identified in the history of AI as an early successful knowledge-intensive expert system. R1/XCON was a later successful commercial expert system, not the first.
Correction: DENDRAL was the first major successful knowledge-intensive system; R1/XCON came later.

الشرح بالعربي: العبارة غلط. DENDRAL هو النظام المعروف كأول نجاح كبير للأنظمة المعتمدة على المعرفة المتخصصة، أما R1/XCON فكان نظامًا ناجحًا ظهر بعده واستخدم في مجال تهيئة أنظمة الحاسوب.

Reference: CS433-W01-Lecture-Notes, §1.3, expert systems era, pp. 13–14.

B6

Statement: MYCIN, which diagnosed blood infections with about 450 rules, handled uncertainty using certainty factors.

Answer: True

MYCIN was a medical expert system for diagnosing blood infections. Its rule-based reasoning used certainty factors to represent degrees of confidence under uncertainty.

الشرح بالعربي: العبارة صح. MYCIN كان نظام خبير لتشخيص عدوى الدم، واعتمد على حوالي 450 قاعدة، واستخدم Certainty Factors عشان يعبر عن درجة الثقة في الاستنتاجات بدل ما يتعامل مع كل معلومة كأنها مؤكدة تمامًا.

Reference: CS433-W01-Lecture-Notes, §1.3, expert systems, pp. 13–14.

B7

Statement: According to the AI Index figures in the book, object-detection error rose from 2% in 2010 to 28% in 2017.

Answer: False

The direction is reversed. The lecture reports that object-detection error fell from 28% in 2010 to 2% in 2017.
Correction: Object-detection error fell from 28% (2010) to 2% (2017).

الشرح بالعربي: العبارة غلط، نسبة الخطأ قلت مش زادت. كانت 28% سنة 2010، وانخفضت إلى 2% سنة 2017، وده معناه إن أداء أنظمة اكتشاف الأشياء اتحسن بشكل كبير.

Reference: CS433-W01-Lecture-Notes, §1.4, state of the art, p. 14.

B8

Statement: Lethal autonomous weapons are a risk because they scale: without human supervision, a small group can deploy any number of weapons.

Answer: True

The lecture identifies scalability as a central risk: autonomous weapons can potentially be deployed at large scale by a small number of people without direct human supervision.

الشرح بالعربي: العبارة صح. خطورة الأسلحة ذاتية التشغيل إنها قابلة للتوسع، يعني عدد قليل من الأشخاص ممكن يشغّلوا أعدادًا كبيرة من الأسلحة من غير إشراف بشري مباشر على كل سلاح.

Reference: CS433-W01-Lecture-Notes, §1.5, near-term risks, p. 16.

B9

Statement: The book's answer to the long-term control problem is to stop AI research.

Answer: False

The lecture explicitly says that the book's answer is not to stop AI research. Instead, machines should pursue human objectives while remaining uncertain about what those objectives are, allowing them to ask, defer, and remain under human control.
Correction: The proposed answer is to design beneficial machines that remain uncertain about human objectives and defer to humans, not to stop AI research.

الشرح بالعربي: العبارة غلط. الحل المقترح مش إيقاف أبحاث الذكاء الاصطناعي؛ الحل هو تصميم أنظمة تسعى لتحقيق أهداف البشر، لكنها تدرك إنها مش عارفة تفضيلاتنا بالكامل، وبالتالي تسأل وتطلب التوجيه وتخضع للتحكم البشري.

Reference: CS433-W01-Lecture-Notes, §1.5, long-term control problem, p. 17.

B10

Statement: In the lab, the brute-force search over weights and thresholds in range(-3, 4) finds no single unit that computes XOR.

Answer: True

The lab code tests all weights and thresholds in the range -3 through 3 and finds an empty set of solutions. This illustrates that a single McCulloch–Pitts unit cannot represent XOR.

الشرح بالعربي: العبارة صح. التجربة بتجرب كل قيم الأوزان والـ Threshold من -3 لحد 3، ومابتلاقيش أي وحدة واحدة تحقق XOR. وده بيوضح إن XOR محتاجة أكتر من طبقة أو أكتر من وحدة.

Reference: CS433-W01-Lecture-Notes, McCulloch–Pitts lab activity, pp. 17–18.

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3 Final Answer Key

Part A – MCQ

A1	C	A2	A	A3 D
A4	B	A5	A	A6 C
A7	D	A8	Second choice*	A9 C
A10	A	A11	D	A12 B
A13	B	A14	C	A15 A
A16	D	A17	C	A18 A
A19	B	A20	D	A21 A
A22	C	A23	B	A24 D
A25	C	A26	A	A27 D
A28	B	A29	C	A30 A

*A8: The original bank duplicates the label (a) for its first and second choices. The correct answer is the second listed choice.

Part B – True / False

B1	F	B2	T	B3	T	B4	F	B5	F
B6	T	B7	F	B8	T	B9	F	B10	T

Important compilation note

Compile this document in Overleaf using **XeLaTeX**. The document uses the Amiri Arabic font and `polyglossia` for Arabic/English support.